

Advanced Statistics for Finance

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1. Upload libraries and data in R

```
1 #Required libraries
2 library(ISLR2)
3 library(e1071)      #for skewness
4 library(leaps)      #for subset selection
5 library(boot)       #for cross-validation
6 library(class)      #for knn
7 library(caret)      #optional for CV and splitting
8 set.seed(2025)      #for reproducibility
9
10 #Load the Data
11 data(Weekly)
12 help(Weekly)
13 str(Weekly)
14 summary(Weekly)
```

2. Data cleaning process

```
15 #Check missing values
16 sum(is.na(Weekly))
17 colSums(is.na(Weekly))
18 #Remove missing values
19 Weekly_clean <- na.omit(Weekly)
20 #% of missing data
21 percentage_missing <- (nrow(Weekly) - nrow(Weekly_clean)) / nrow(Weekly) * 100
22 cat("Percentage of missing data:", percentage_missing, "%\n")
```

```
> sum(is.na(Weekly))
[1] 0
> colSums(is.na(Weekly))
      Year      Lag1      Lag2      Lag3      Lag4      Lag5      Volume      Today      Direction
      0         0         0         0         0         0         0         0         0
> #Remove missing values
> Weekly_clean <- na.omit(Weekly)
> #% of missing data
> percentage_missing <- (nrow(Weekly) - nrow(Weekly_clean)) / nrow(Weekly) * 100
> cat("Percentage of missing data:", percentage_missing, "%\n")
Percentage of missing data: 0 %
```

These are the results of my data cleaning process, no missing data were found in the Weekly dataset, therefore no rows were removed.

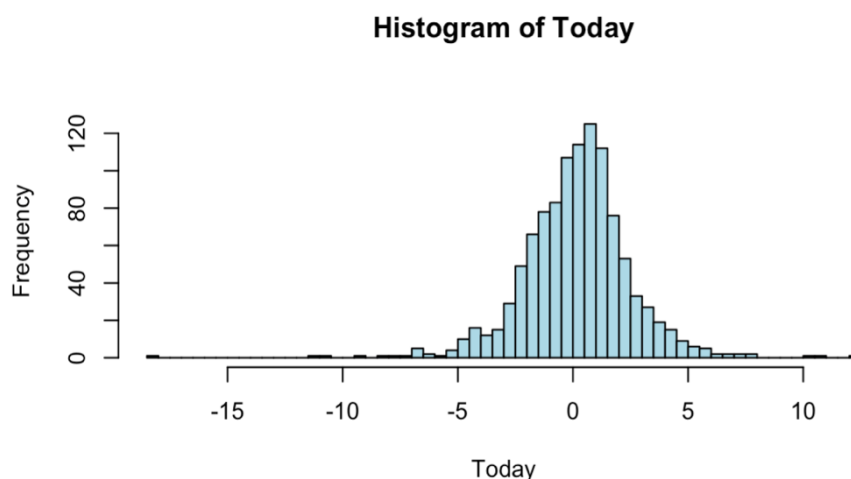
3. Descriptive analysis of the variable ‘Today’

```
24 #Descriptive Analysis of 'Today': Mean, SD, Skewness, Kurtosis, Median and Histogram
25 mean_today <- mean(Weekly_clean$Today)
26 sd_today <- sd(Weekly_clean$Today)
27 skew_today <- skewness(Weekly_clean$Today)
28 kurt_today <- kurtosis(Weekly_clean$Today)
29 median_today <- median(Weekly_clean$Today)
30 cat("Mean of Today:", mean_today, "\n")
31 cat("SD of Today:", sd_today, "\n")
32 cat("Skewness of Today:", skew_today, "\n")
33 cat("Kurtosis of Today:", kurt_today, "\n")
34 cat("Median of Today:", median_today, "\n")
35 hist(Weekly_clean$Today, breaks=50, main="Histogram of Today", xlab="Today", col = "lightblue")
```

From the analysis, i've obtained the following results:

- Mean = 0.149899
- Median = 0.241
- Standard Deviation = 2.356927
- Kurtosis = 5.669553 → higher than the value 3 expected for a normal one; presence of heavy tails in the distribution of weekly returns, i.e. a greater frequency of extreme events. This result is consistent with the known behavior of financial markets, which assume anomalous variations with a higher probability than what happens with a normal distribution.
- Skewness = -0.4798404 → slight negative skewness of the weekly returns distribution.

Each result obtained through the descriptive analysis is proved by the following figure, showing a distribution centered around 0.



4. Multiple linear regression analysis

$$\text{Today}_i = \beta_0 + \beta_1 \text{Year}_i + \beta_2 \text{Lag1}_i + \beta_3 \text{Lag2}_i + \beta_4 \text{Lag3}_i + \beta_5 \text{Lag4}_i + \beta_6 \text{Lag5}_i + \beta_7 \text{Volume}_i + \epsilon_i$$

This is the multiple linear model that i've considered, intentionally leaving out the variable 'Direction', because it denotes with Up and Down whether the market had a positive or negative return on that week, so it's just a categorical transformation of Today that can lead to a leakage problem. Including a variable that is derived from the response variable generates a model with a higher Adjusted R^2 , but this doesn't match the reality. I did not maintain it into the linear model

because even the forward and backward selections would have been affected by it, distorting the explanatory power of the model. The variable ‘Year’, denoting the years from 1990 to 2010, was considered; i’ve done this choice because it can be excluded later by the variable selection methods.

```
40 #Linear Regression taking variable 'Today' as Response
41 lm_model <- lm(Today ~ . - Direction, data = Weekly_clean)
42 summary(lm_model)
```

```
Call:
lm(formula = Today ~ . - Direction, data = Weekly_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-18.5595  -1.2597   0.0706   1.2635  11.9010

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.627829  43.922572   0.174   0.8622
Year          -0.003698   0.022014  -0.168   0.8666
Lag1          -0.071183   0.030505  -2.333   0.0198 *
Lag2           0.048017   0.030697   1.564   0.1181
Lag3          -0.068275   0.030519  -2.237   0.0255 *
Lag4          -0.021925   0.030519  -0.718   0.4727
Lag5           0.014276   0.030434   0.469   0.6391
Volume        -0.043135   0.079586  -0.542   0.5879
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.347 on 1081 degrees of freedom
Multiple R-squared:  0.01471,    Adjusted R-squared:  0.008332
F-statistic: 2.306 on 7 and 1081 DF,  p-value: 0.02459
```

4.1 Commented results

The main findings are the following:

- The model has an Adjusted R^2 equal to 0.0083, meaning that it explains less than 1% of the variability of Today. This result is expected, given the extremely noisy nature of weekly stock market returns, which tend to behave like a random walk and the fact that past returns badly explain future returns.
- The Residual Standar Error is 2.347 on 1081 degrees of freedom, coherent with the low value of the Adjusted R^2 .
- The intercept estimate is more or less 7.63; this means when the other variables are 0, the variable Today assumes that value. This result is expected as well, because in the dataset the value 0 for the other variables (especially Year and Volume) is not realistic. So it is not significant, as proved by the Standard error of the intercept that is very huge (43.9).
- Focusing on the independent variables, only ‘Lag1’ and ‘Lag3’ show a statistically significant relationship with current returns, even though that relationship is very weak. This suggests the presence of a weak mean reversion effect in the weekly return of the S&P 500 Index: positive returns in the previous weeks tend to be followed by slightly negative returns and viceversa (because the estimate is negative). However, the effect is very small and the overall model explains only a small part of the variability in returns.

The other variables do not make a significant contribution to Today's forecast, consistent with the financial theory and the idea that weekly returns are difficult to be predicted based on simple historical variables.

5. Variable selection and model choice processes

In statistics one of the main problem is linked to the choice of the best subset of predictors to use for finding the model that best explains the response variable. In general, models with a less number of variables are preferred, because easier to interpret, for avoiding overfitting and so on and so forth. The most used methods for the variable selection are the stepwise methods, while BIC, AIC, Adjusted R^2 are used for choosing the best model.

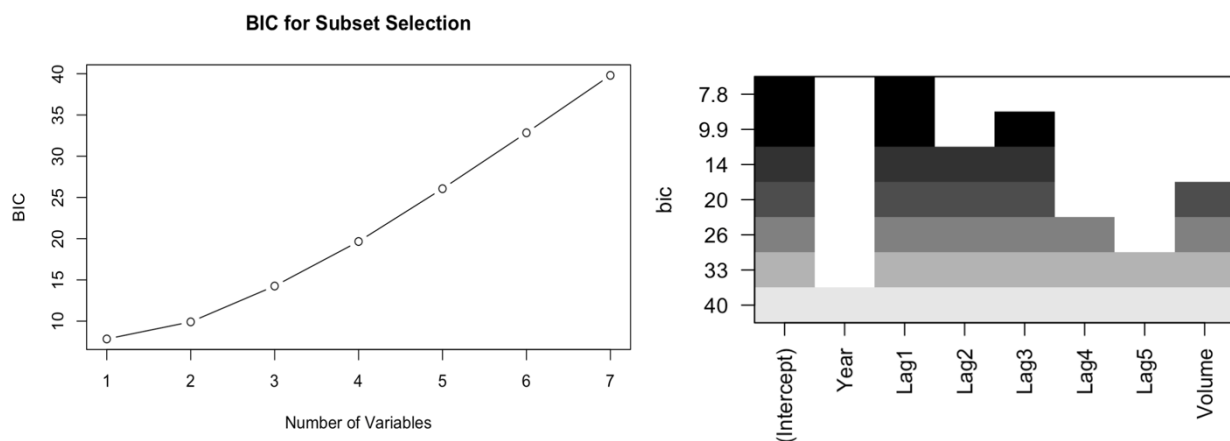
5.1 BIC and Stepwise selection methods: implementation and results

Fist of all, i've computed the BIC for each combination of predictors, which penalizes models with too much variables, seeking simple models; the lower the BIC the higher is the performance of the model. Then, I've used both the forward and the backward stepwise method, focused on the BIC values; the difference is that the first one starts from an empty model and adds step by step the variables giving the highest additional improvement to the model, while the second one begins with the full model, removing the most useless variables.

```

43 #Best Subset Selection
44 regfit.full <- regsubsets(Today ~ . - Direction, data = Weekly_clean, nvmax = 7)
45 reg_summary <- summary(regfit.full)
46 names(reg_summary)
47 #Plot BIC
48 reg_summary$bic
49 plot(reg_summary$bic, main="BIC for Subset Selection", xlab="Number of Variables", ylab="BIC", type = "b")
50
51 regfit.fwd=regsubsets(Today ~ . - Direction, data = Weekly_clean,
52                       nvmax=7,method="forward")
53 summary(regfit.fwd)
54 reg.summary=summary(regfit.fwd)
55 which.min(reg.summary$bic)
56 points(11,reg.summary$bic[11],pch=20)
57 plot(regfit.fwd,scale="bic")
58
59 regfit.back=regsubsets(Today ~ . - Direction, data = Weekly_clean,
60                       nvmax=7,method="backward")
61 summary(regfit.back)
62 reg.summaryb=summary(regfit.back)
63 which.min(reg.summaryb$bic)
64 points(11,reg.summaryb$bic[11],pch=20)
65 plot(regfit.back,scale="bic")

```



The previous figures show that the BIC is lower when only one variable is considered, and then increases when we add other variables, so the stepwise selection methods lead to the same result, suggesting a very simple model made of only one variable, that is 'Lag1'.

The next step is to implement the two models suggested before, look at the results and compare them with the original model:

```
67 forward_model <- lm(Today ~ Lag1, data = Weekly_clean)
68 summary(forward_model)
69 backward_model <- lm(Today ~ Lag1, data = Weekly_clean)
70 summary(backward_model)
```

```
Call:
lm(formula = Today ~ Lag1, data = Weekly_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-19.061  -1.271   0.113   1.280  11.235

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.16120    0.07140   2.258  0.0242 *
Lag1        -0.07503    0.03024  -2.481  0.0133 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.351 on 1087 degrees of freedom
Multiple R-squared:  0.00563,    Adjusted R-squared:  0.004715
F-statistic: 6.154 on 1 and 1087 DF,  p-value: 0.01326
```

The selection methods have suggested that a model with a single predictor variable, 'Lag1', is optimal, based on the BIC. In fact, the BIC reaches its minimum value in the model with a single variable, indicating that the addition of predictors does not justify the penalty due to the increase in complexity. However, in the transition from the complete model (which includes all available variables except 'Direction') to the simplified model with only 'Lag1', a decrease in the Adjusted R^2 is observed: it goes from a value of approximately 0.0083 to 0.0047. This means that, in terms of explanatory capacity on the training data, the simpler model explains an even smaller part of the variability of the target variable 'Today'. Although the other variables marginally improve the model fit to the observed data, they do not provide enough contribution to justify their inclusion in a more efficient predictive model. A slightly lower Adjusted R^2 may be acceptable if in exchange it results in a much simpler model (more interpretable, less noise...).

The final model has a very low Adjusted R^2 as well, highlighting that:

- the weekly returns are highly unpredictable in the short term;
- the relationship between past and future returns is very weak;
- the observed effects are low and do not allow an accurate forecast of Today.

5.2 K-fold Cross Validation: differences and final model choice

The last method used is the K-fold Cross Validation with $k = 10$; this approach divides the entire dataset into 10 groups and each group will take the role of the test set, while the other $k-1$ will be considered as a training set. At the end it does the mean of all the ten values of the prediction error computed for each test group.

I've implemented the k-fold CV for both the models, with both glm and caret, obtaining:

1) First model:

a) 10-fold CV Error: 5.640663 5.631395

b) RMSE = 2.351987, Rsquared = 0.01375293, MAE = 1.6962

2) Second model:

a) 10-fold CV Error: 5.545656 5.54424

b) RMSE = 2.338929 Rsquared = 0.0306128, MAE = 1.688354

The value obtained, using caret, for the RMSE coincides with the value obtained for the RSE coming from the model created by the forward and the backward analysis.

The most important aspect to focus on is the partial decrease of the CV Error going from the first model to the second one, meaning that the model with only one covariate better generalizes.

The simplified model has a lower predictive error, proving that, although the full model fits the training data better (greater Adjusted), the risk of overfitting is higher. Therefore, the second model is preferable from a forecasting perspective, favoring generalization rather than complexity. This situation reflects a fundamental principle of statistics: a more complex model is not necessarily the best one, especially if the goal is not only to describe the data but also to predict new observations. The best compromise between simplicity and predictive capacity, in this case, turns out to be the model with a single variable, as indicated by stepwise selection methods and confirmed by cross-validation. Mixing everything, there are some differences between the models, and the final choice depends on the original goal: the one proposed by the stepwise methods is better if the goal is the prediction, instead the first one is recommended for interpretation tasks.

6. K-NN analysis on 'Direction'

The k-NN (k-Nearest Neighbors) is one of the most easy and intuitive methods used in Machine Learning and Data mining, for both classification and regression goals.

'Direction', which is a binary variable (up and down), has been chosen as prediction variable; therefore in this case, its purpose is to understand whether it is possible to predict the future direction of the S&P500 index based on some other features.

In the k-NN classification phase, to simulate a realistic application (for example for a weekly trading system), is important to use only information that would be known before the start of the week. The variable 'Today', which represents the percentage return of the current week, is directly linked to 'Direction' ('Direction' is determined by Today: UP if 'Today' > 0, DOWN if 'Today' < 0). Including 'Today' among the predictors would have introduced a data leakage problem (look-ahead bias), that is, the use of information that are not available at the time of the forecast and that make the evaluation of the model incorrect.

For this reason, I used as input only the percentage returns of the previous weeks ('Lag1',... 'Lag5'), and 'Volume', which represent historical information actually known at the time of the forecast, and the variable Year which does not change too much the calculation of the Euclidean distances, the confusion matrix and the accuracy.

```
99 #Apply k-NN with k=3 (i've also tried with different values)
100 knn_pred <- knn(train=train_X, test=test_X, cl=train_Y, k=5)
101 #Confusion matrix creation
102 confusion_matrix <- table(knn_pred, test_Y)
103 print(confusion_matrix)
104 #Calculate accuracy
105 accuracy <- mean(knn_pred == test_Y)
106 cat("k-NN Accuracy:", accuracy, "\n")
```

After splitted data in Training set (70%) and Testing set (30%) and added other details, the previous code led to the following outcomes:

1)

test_Y		
knn_pred	Down	Up
Down	65	54
Up	89	119

This is the confusion matrix, indicating how many times the model has either correctly or wrongly predicted.

- 2) k-NN Accuracy: 0.5626911 → 56%; this means that the model has been accurate for the 56%, a result moderately above the 50% chance level. This indicates that the information contained in past returns ('Lag1'–'Lag5') and 'Volume' have a modest predictive power on the weekly direction of the market. Even though it's not a high accuracy, in an environment such as financial markets -- notoriously difficult to predict -- even a small increase can be significant. The result, nevertheless, confirms that forecasting the short-term direction remains complex and subject to strong variability.